

BLUE TEAM OPERATOR 1 LAB TRACK

T Ngwako

Contents

Introduction.....	2
Objectives	2
Step 1 - Signal Onboarding and health.	2
Step 2 - Windows Security via AMA	5
Step 3 - Sysmon +DCR	6
KQL Essentials	8
Step 4 -OOB rule creation.....	9
Step 5 – Workbook 101	10
Step 6 – workbook 101 – import high value users	12
Conclusion	14

Introduction

This lab track documents the process of onboarding a Windows virtual machine (VM) into Microsoft Sentinel and validating its security signal health. As part of the Blue Team Operator Level 1 role, the lab focused on confirming that key logs such as Heartbeat, SecurityEvent, and Sysmon were being collected correctly. It also included installing and configuring Azure Monitor Agent (AMA), setting up Data Collection Rules (DCR), enabling built-in detection rules, building visual workbooks, and importing a watchlist to monitor high-value users. Each step was completed to ensure visibility, detection coverage, and enrichment capabilities within the Sentinel environment.

Objectives

1. Confirm that Heartbeat and SecurityEvent logs are landing from the VM
2. Install and verify Azure Monitor Agent (AMA) for Windows Security log collection
3. Deploy Sysmon with a modular configuration and validate process creation events
4. Use KQL to test log freshness, event IDs, and basic filtering and summarization
5. Enable out-of-the-box (OOB) detection rules from Sentinel's rule templates
6. Build workbooks to visualize login activity and Sysmon process execution
7. Import and test a watchlist for tracking high-value users or assets

Step 1 – Signal Onboarding and Health

Purpose: Ensures that Microsoft Sentinel is properly connected to the VM and actively receiving logs. Confirms log freshness using Heartbeat and SecurityEvent data, and build a health workbook to visualize signal status.

Steps taken:

1. Navigated to Sentinel → workspace → Logs.

Ran this query to check Heartbeat freshness (shows when your agents last reported):

Heartbeat

| summarize LastHeartbeat = max(TimeGenerated) by Computer

| extend AgeInMinutes = (datetime_diff('minute', now(), LastHeartbeat))

| sort by AgeInMinutes asc

Selected workspace: 't1-law'

New Query 1* ... X +

Save Share ...

Run Time range: Last 24 hours Show: 1000 results

```
1 Heartbeat
2 | summarize LastHeartbeat = max(TimeGenerated) by Computer
3 | extend AgeInMinutes = (datetime_diff('minute', now(), LastHeartbeat))
4 | sort by AgeInMinutes asc
5
```

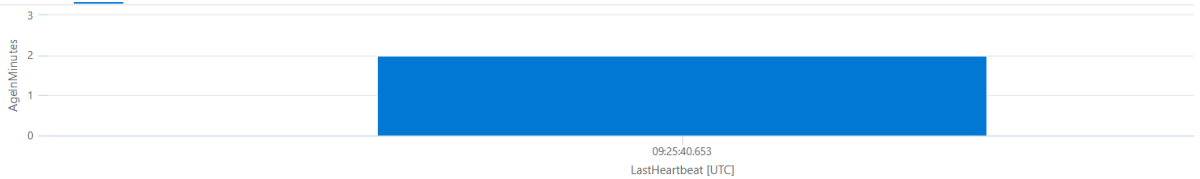
Results Chart Add bookmark

Computer	LastHeartbeat [UTC]	AgeInMinutes
> NODE02-T1	06/10/2025, 09:25:40.653	2

Run Time range: Last 24 hours Show: 1000 results

```
1 Heartbeat
2 | summarize LastHeartbeat = max(TimeGenerated) by Computer
3 | extend AgeInMinutes = (datetime_diff('minute', now(), LastHeartbeat))
4 | sort by AgeInMinutes asc
5
```

Results Chart



● NODE02-T1

2. Checked Security event freshness by running the following query:
SecurityEvent
| summarize LastEvent = max(TimeGenerated) by Computer
| extend AgeInMinutes = datetime_diff('minute', now(), LastEvent)
| sort by AgeInMinutes asc

Selected workspace: 't1-law'

New Query 1* ... X +

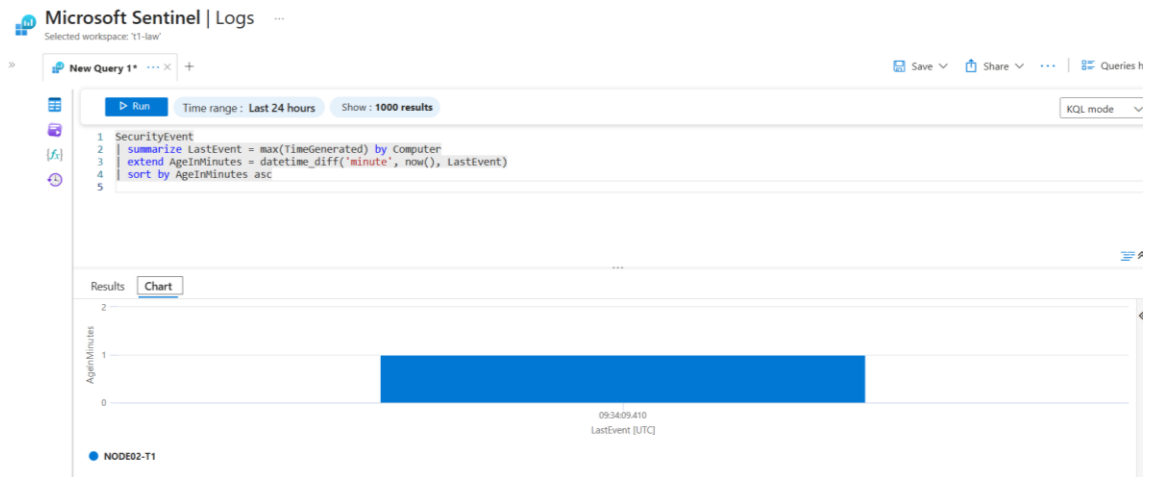
Save Share ... Queries hub

Run Time range: Last 24 hours Show: 1000 results KQL mode

```
1 SecurityEvent
2 | summarize LastEvent = max(TimeGenerated) by Computer
3 | extend AgeInMinutes = datetime_diff('minute', now(), LastEvent)
4 | sort by AgeInMinutes asc
5
```

Results Chart Add bookmark

Computer	LastEvent [UTC]	AgeInMinutes
> NODE02-T1	06/10/2025, 09:34:09.410	1



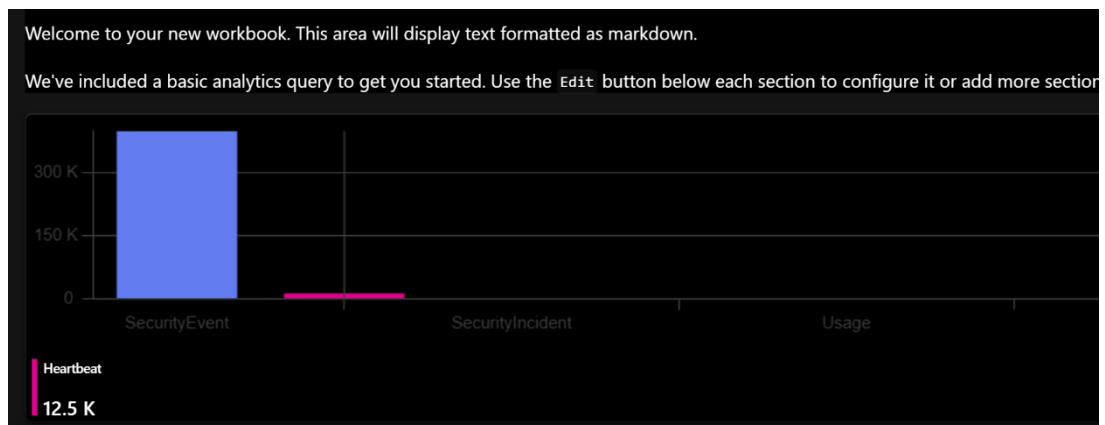
AgeInMinutes is 1, meaning the most recent SecurityEvent from the VM was received one minute ago, indicating that log ingestion is healthy and up to date.

3. Creating a Health Workbook

A health workbook helps visualize signal freshness and data ingestion.

In Microsoft Sentinel, navigated to → Workbooks → click + Add workbook.

There should be a Health Workbook that displays both SecurityEvent and Heartbeat data. If this workbook is already available, it confirms that Sentinel is properly collecting and displaying data from connected sources.



However, if the workbook is not present, it can be created manually by following these steps:

Open Workbooks

- In Microsoft Sentinel, select your workspace.
- Go to Workbooks → click + Add workbook.

Add a Heartbeat query

- Click Edit → then Add query.
- Paste the following query to monitor agent connectivity:

```

• union withsource=_TableName *
• | summarize Count=count() by _TableName
• | render barchart

```

The query will return a chart with Security Event, usage, security alert, and anomalies.

Step 2 – Windows Security via AMA

Purpose: collect essential Windows Security logs from the VM using Azure Monitor Agent (AMA). This includes logon success, logon failure, and process creation events. Confirming Event IDs 4624, 4625, and 4688 ensures that key security signals are being captured.

Steps taken:

1. Installed Azure Monitor Agent (AMA) on the VM:
 - Azure Portal → Sentinel → Content Hub → Searched Windows Security Event → Selected Windows Security Event → Clicked Install.
2. Verified AMA was installed on the VM:
 - Azure Portal → Virtual Machines → VM → Extensions.

Home > NODE02-T1

NODE02-T1 | Extensions + applications

Virtual machine

Search to filter items...

Showing all 1 items

Name	Type	Version	Latest Version	Status	Automatic upgrade status
AzureMonitorWindowsAgent	Microsoft.Azure.Monitor...	1.38.1.0	1.38.1.0	Provisioning succeeded	Disabled

3. Searched Data Collection Rules on Azure portal, selected it, and clicked + Create.
4. Entered DCR name (DCR-Windows-NODE02), selected subscription and resource group, and linked the VM.
5. Saved and applied the DCR.
6. Verified Collection Rule was created, searched Data Collection Rule on Azure

Microsoft Azure

Home > Data collection rules

Default Directory

Filter for any field...

Name	Subscription	Resource Group	Location	Data sources	Destinations	Kind
DCR-AMA	Azure subscription 1	Tier1_SOC	South Africa North	Windows Event Logs	Azure Monitor Logs	Windows
DCR-Windows-NODE02	Azure subscription 1	NODE02-T1_group	South Africa North	Windows Event Logs	Azure Monitor Logs	Windows
DCR_collector	Azure subscription 1	Tier1_SOC	South Africa North	Windows Event Logs	Azure Monitor Logs	Windows

7. Confirmed EventIDs 4624/4625/4688 by running:

```
SecurityEvent
| where EventID in (4624, 4625, 4688)
| summarize Count = count() by EventID
```

▶ Run Time range : Custom Show : 1000 results

```
1 SecurityEvent
2 | where EventID in (4624, 4625, 4688)
3 | summarize Count = count() by EventID
```

Results Chart Add bookmark

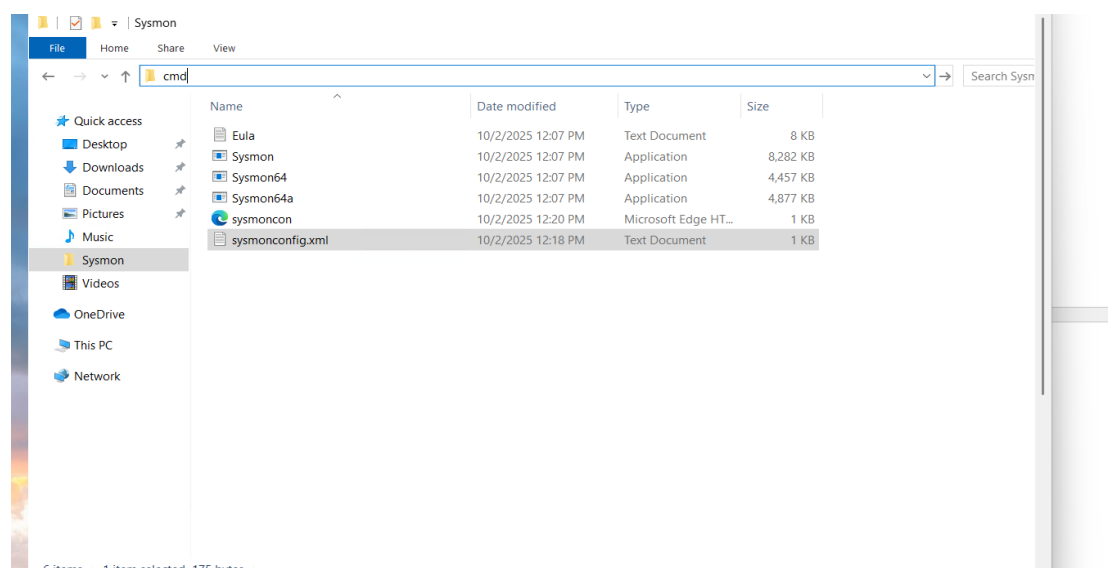
EventID	Count
> 4625	382259
> 4688	13496
> 4624	2003

Step 3 – Sysmon + DCR

Purpose: To enhance visibility into system-level activity by installing Sysmon with a curated configuration. A Data Collection Rule (DCR) is used to collect logs from the Sysmon Operational channel, allowing detection of detailed process and behavior data.

Steps taken:

1. Within the virtual Machine, downloaded Sysmon from the following link: [Sysmon - Sysinternals | Microsoft Learn](#)
2. Unzipped the folder, then within unzipped folder, pasted an xml file downloaded from the link [sysmon-modular/sysmonconfig.xml at master · olafhartong/sysmon-modular · GitHub](#)
3. Open the Sysmon directory in cmd



4. Ran the following command to install sysmon:

sysmon -accepteula -i sysmonconfig.xml

5. Confirmed Event logging by navigating to Event Viewer → Application and Services logs → Microsoft → Windows → Sysmon → Operational.

Level	Date and Time	Source	Event ID	Task Category
Information	10/8/2025 7:08:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:31 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	10/8/2025 7:08:31 AM	Sysmon	1	Process Create (rule: ProcessCre...
Information	10/8/2025 7:08:31 AM	Sysmon	7	Image loaded (rule: ImageLoad)

6. Navigated to Azure and Search Data Collection Rules → +Create

- Under Basics, entered rule name, choose resource group with the VM, choose region.
- Under Resources, click Add resource → choose appropriate resource group with VM → Apply
- Under Collect and deliver, +Add Data Source, Selected Windows Event Logs as Data Source type, click on customs, then add the following query in the xpath box

```
<QueryList>
```

```
<Query Id="0" Path="Microsoft-Windows-Sysmon/Operational">
```

```
<Select Path="Microsoft-Windows-Sysmon/Operational">*</Select>
```

```
</Query>
```

```
</QueryList>
```

Click Add after pasting the query

Under Destination still within Collect and deliver, selected Azure Monitor logs as destination type, select your said subscription under subscription, select your said Log Analytics Workspace under Destination details.

Click Add Data Source

- Clicked Review and Create

7. Confirm Logging

Navigated to Log Analytics Workspace → Logs and paste the following KQL query
Event
| where Source == "Microsoft-Windows-Sysmon"

Query editor						
Results Chart						
TimeGenerated [UTC] ↑↓	Source	EventLog	Computer	EventLevel	EventLevelName	ParameterXml
> 07/10/2025, 13:26:13.529	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/Operational	NODE02-T1	4	Information	<Param> technique_id=T1021,technique_name=Remote Ser...
> 07/10/2025, 13:26:09.670	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/Operational	NODE02-T1	4	Information	<Param> technique_id=T1021,technique_name=Remote Ser...
> 07/10/2025, 13:26:01.904	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/Operational	NODE02-T1	4	Information	<Param> technique_id=T1021,technique_name=Remote Ser...
> 07/10/2025, 13:26:01.655	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/Operational	NODE02-T1	4	Information	<Param> -</Param> <Param> SetValue</Param> <Param> 2...
> 07/10/2025, 13:25:59.873	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/Operational	NODE02-T1	4	Information	<Param> technique_id=T1021,technique_name=Remote Ser...
> 07/10/2025, 13:25:58.123	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/Operational	NODE02-T1	4	Information	<Param> technique_id=T1574.010,technique_name=Service...
> 07/10/2025, 13:25:58.060	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/Operational	NODE02-T1	4	Information	<Param> technique_id=T1021,technique_name=Remote Ser...
> 07/10/2025, 13:25:54.170	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/Operational	NODE02-T1	4	Information	<Param> technique_id=T1021,technique_name=Remote Ser...
> 07/10/2025, 13:25:47.732	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/Operational	NODE02-T1	4	Information	<Param> technique_id=T1021,technique_name=Remote Ser...
> 07/10/2025, 13:25:46.951	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/Operational	NODE02-T1	4	Information	<Param> technique_id=T1021,technique_name=Remote Ser...
> 07/10/2025, 13:25:45.654	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/Operational	NODE02-T1	4	Information	<Param> -</Param> <Param> SetValue</Param> <Param> 2...

Step 4 – KQL Essentials

Purpose: To build foundational skills in Kusto Query Language (KQL). This step focuses on learning how to filter, group, and analyze data using basic operators like where, summarize, and bin. These queries are essential for threat hunting and workbook creation.

Steps taken:

1. Navigated to Log Analytics workspace → Logs and paste the following queries:

a) Where

Filters the data to show only the rows that meet a condition.

Example: Records when a user session ends

SecurityEvent

| where EventID == 4634

b) Summarize

Groups and aggregates data, for example, counting how many times something happened.

Common aggregation functions: count(), avg(), sum(), max(), etc.

Example: count failed logins per user.

SecurityEvent

| where EventID == 4625

| summarize FailedLogins = count() by Account

Run Time range: Last 3 days Show: 1000 results KQL mode

Query editor

Results Chart

Account	FailedLogins
> \BRR	2
> \ADMIN	4506
> \FFF	2
> \USER	4393
> \VW	2
> \TTT	2
> \GGG	2
> \BBB	2
> \YYY	2
> \HHH	2
> \NNN	2

c) Bin

Used with summarize to group data into time intervals.

Helps create trend charts or visualize data over time.

Example: count failed logins per 1-hour period.

SecurityEvent

| *where EventID == 4625*

| *summarize FailedLogins = count() by bin(TimeGenerated, 1h)*

Run Time range: Last 3 days Show: 1000 results

```

1 SecurityEvent
2 | where EventID == 4625
3 | summarize FailedLogins = count() by bin(TimeGenerated, 1h)

```

Results Chart

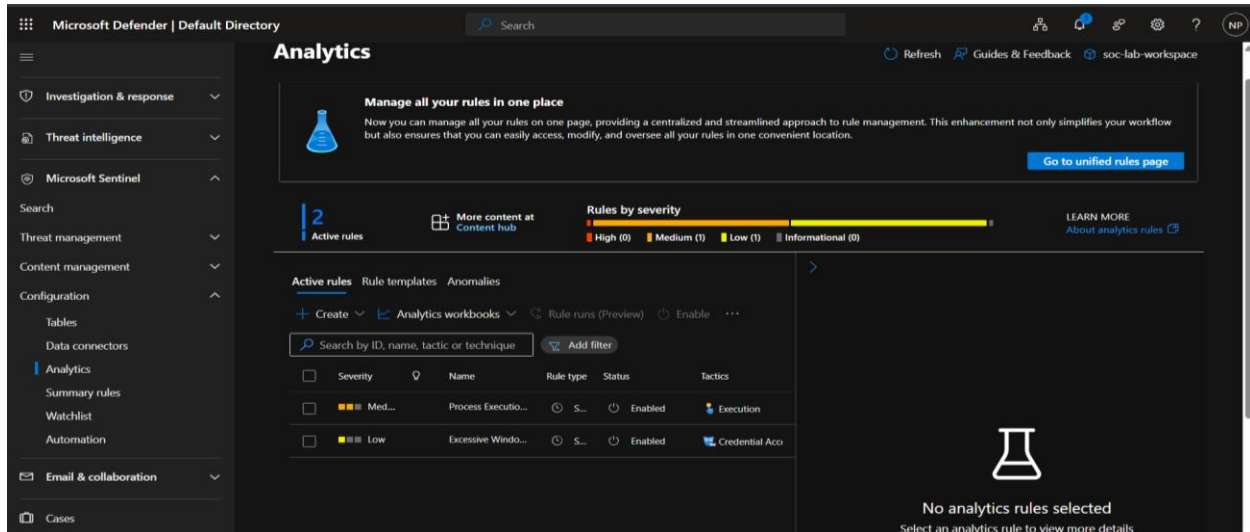
TimeGenerated [UTC] ↑↓	FailedLogins
> 07/10/2025, 08:00:00.000	542
> 07/10/2025, 07:00:00.000	681
> 06/10/2025, 13:00:00.000	1005
> 06/10/2025, 12:00:00.000	1018
> 06/10/2025, 11:00:00.000	525
> 06/10/2025, 10:00:00.000	459
> 06/10/2025, 09:00:00.000	338
> 06/10/2025, 08:00:00.000	352

Step 5 – OOB Rule Creation

Purpose: To activate built-in analytics rules in Sentinel that detect common threats such as brute-force attacks and suspicious processes. Enabling these rules helps automate detection and generate alerts based on known attack patterns.

Steps taken:

1. Went to Sentinel → Configuration → Analytics → + Rule Template
2. Selected “Excessive Windows Logon Failures” → Create rule → Review and create → save.
3. Selected “Process Execution Frequency Anomaly” → Create rule → Review and create → save.



Step 6 – Workbook 101

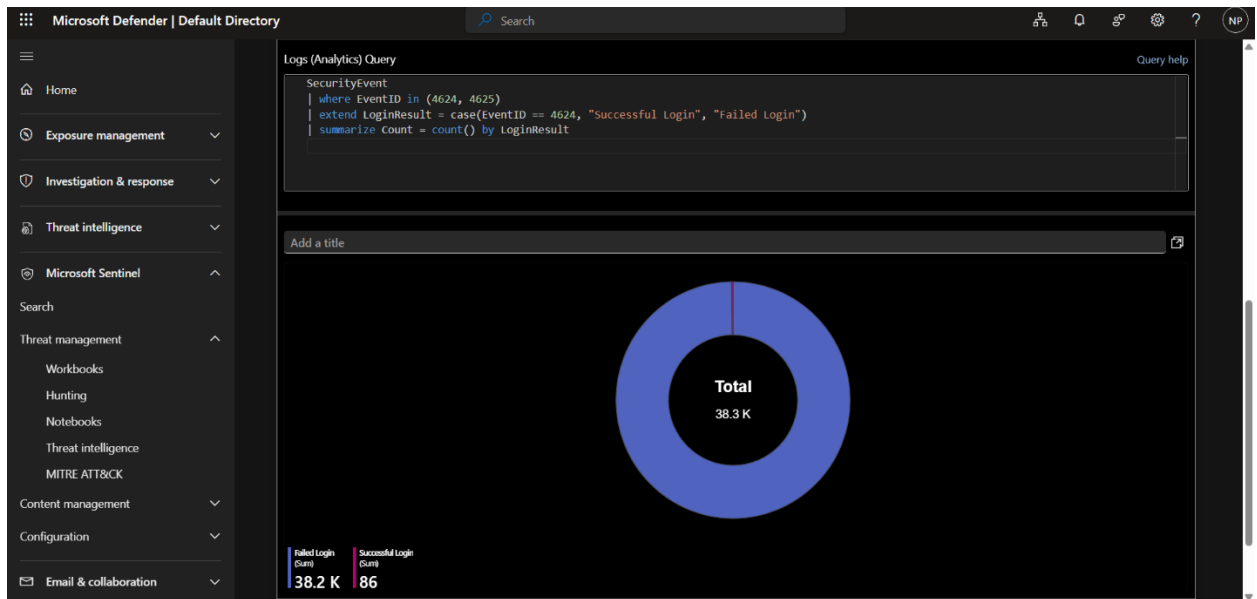
Purpose: To create visual dashboards that display key security data. This includes charts for failed vs successful logons and top N Sysmon processes. Workbooks help analysts quickly understand trends and spot anomalies.

Steps taken:

1. Went to Microsoft Sentinel → Workbooks → + New Workbook → Edit → +Add
2. Added Query control, pasted the following Logons query:

```
SecurityEvent
| where EventID in (4624, 4625)
| extend LoginResult = case(EventID == 4624, "Successful Login", "Failed Login")
| summarize Count = count() by LoginResult
```

3. Chose a Pie Chart for visual formatting.
4. Clicked Run Query.



1. For Sysmon process top-N, clicked + New Workbook → Edit → +Add Queried Sysmon process creation events using the following KQL query:

Event

| where Source == "Microsoft-Windows-Sysmon" and EventID == 1

| extend Image = tostring(EventData.Image)

| summarize Count = count() by Image

| top 10 by Count

Query failed

Run Time range : Last 7 days Show : 1000 results

```

1 Event
2 | where Source == "Microsoft-Windows-Sysmon" and EventID == 1
3 | extend Image = tostring(EventData.Image)
4 | summarize Count = count() by Image
5 | top 10 by Count
6
```

Results Chart

❌ Path expression Image source must be scalar of type 'dynamic'. Received a source of type string instead
Request id: 100dc72e-1135-41eb-a314-7fe5fac583c9

2. Attempted alternative approaches using the Message field, such as:

Event

| where Source == "Microsoft-Windows-Sysmon" and EventID == 1

| extend ImagePath = extract(@"Image:\s*([^\r\n]*)", 1, Message)

| summarize Count = count() by ImagePath

| top 10 by Count

Query returned a table, but the imagePath column was empty.

The screenshot shows the Azure Sentinel query editor interface. At the top, there's a tab labeled 'New Query 1*'. Below it, a 'Run' button is visible, along with filters for 'Time range: Last 7 days' and 'Show: 1000 results'. The query text is as follows:

```
1 Event
2 where Source == "Microsoft-Windows-Sysmon" and EventID == 1
3 extend ImagePath = extract(@"Image:\s*(^[^r\n]*)", 1, Message)
4 summarize Count = count() by ImagePath
5 top 10 by Count
6
7
```

Below the query editor, there are two tabs: 'Results' and 'Chart'. The 'Results' tab is active, displaying a table with two columns: 'ImagePath' and 'Count'. The table contains one row with the value '>' under 'ImagePath' and '18' under 'Count'.

ImagePath	Count
>	18

This indicates that either:

- The Sysmon configuration on the VM does not log full process paths for EventID 1.
- The logs were not fully ingested into Azure Sentinel.

Expected Results (if logs were complete):

A table of process names (Image or ImagePath) and their execution counts.

The Top-N most frequent processes would be visible, helping identify abnormal or rare process execution.

Step 7 – Watchlist 101 – Import High-Value Users

Purpose: To import and use a list of high-value users or assets for focused monitoring.

Watchlists allow you to enrich queries and detections by highlighting activity related to sensitive or critical accounts.

Steps taken:

Phase 1: Prepare the Watchlist CSV File

Created a small CSV file containing:

- Name - hostname or username you want to track.
- Type - indicates whether it's a user or server.
- Tag - optional classification (Critical, High)

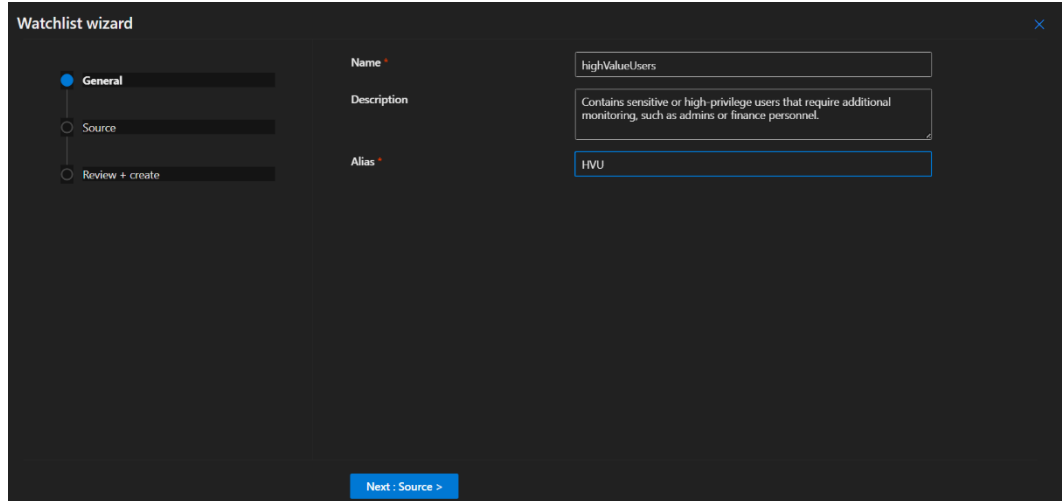
Phase 2: Import the Watchlist into Sentinel

- 2.1. In the Azure portal, navigated to Microsoft Sentinel → Workspace → Configuration → Watchlists.
- 2.2. Clicked + Add new.
- 2.3. Filled out the form:

Name: HighValueAssets

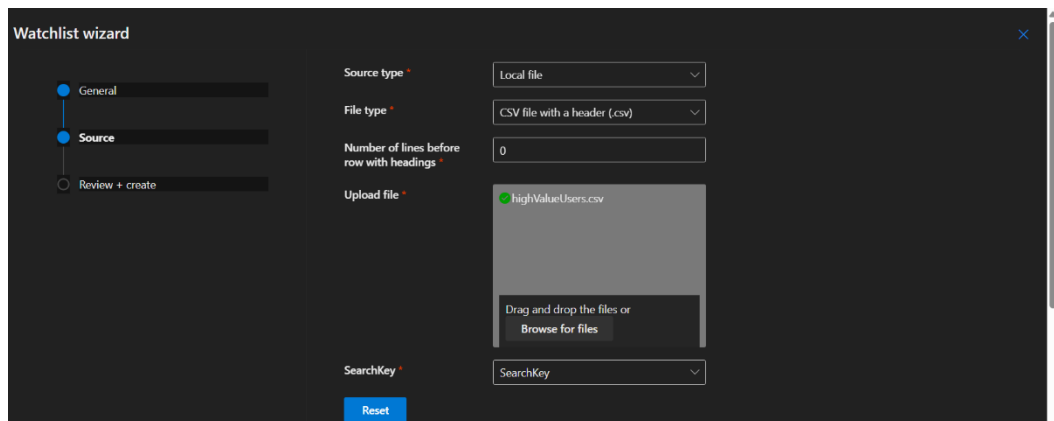
Description: Contains sensitive or high-privilege users that require additional monitoring, such as admins

Alias: highvalue (this is important — used in queries)

The screenshot shows the 'Watchlist wizard' interface with the 'General' tab selected. On the left, there is a vertical navigation bar with three options: 'General' (selected with a blue dot), 'Source', and 'Review + create'. The main area contains three input fields: 'Name' with the value 'highValueUsers', 'Description' with the text 'Contains sensitive or high-privilege users that require additional monitoring, such as admins or finance personnel.', and 'Alias' with the value 'HVV'. At the bottom right, there is a blue button labeled 'Next : Source >'. The interface has a dark theme.

2.4. Clicked next:Source

2.5. Uploaded CSV file

The screenshot shows the 'Watchlist wizard' interface with the 'Source' tab selected. The left navigation bar now has 'Source' selected with a blue dot. The main area contains several configuration options: 'Source type' set to 'Local file', 'File type' set to 'CSV file with a header (.csv)', 'Number of lines before row with headings' set to '0', and 'Upload file' showing a green checkmark and the filename 'highValueUsers.csv'. Below these is a 'Browse for files' button. At the bottom, there is a 'SearchKey' dropdown menu and a blue 'Reset' button. The interface has a dark theme.

2.6. Click Next, then Create.

After a few moments, the Watchlist appeared in the list.

3. Using the Watchlist in a KQL Query.

The watchlist was successfully uploaded to Sentinel and appeared under Configuration → Watchlists; however, when queried using the `_GetWatchlist()` or lookup operator in the Logs workspace, no results were returned.

Possible reasons:

- The watchlist name used in the query did not match the internal watchlist identifier.
- There may have been a workspace scope or permission mismatch.

- The watchlist data may not have been properly indexed yet.

Conclusion

This lab track successfully confirmed that the VM was sending logs to Microsoft Sentinel, with Heartbeat and SecurityEvent data showing recent activity. AMA and Sysmon were installed and configured correctly, and key Event IDs (4624, 4625, 4688) were verified. KQL queries were used to test log freshness and explore login and process activity. Built-in detection rules were enabled, and workbooks were created to visualize key signals. A watchlist was imported to help monitor sensitive users, although initial query issues highlighted the importance of matching aliases and verifying indexing. Overall, the lab demonstrated essential Blue Team Operator Level 1 skills in signal onboarding, detection setup, and data visualization.